

Says Anupama Mohanram, Co-founder and Head of Architecture at Green Evolution, a Chennai-based firm that specialises in building energy efficient and earth-friendly spaces: “Use natural materials like mud blocks or hollow terracotta blocks that are good insulators. Concrete and brick are very poor insulators. They absorb the heat from outside during the day and radiate it inside during the night. Concrete heats up the place; it’s also not sustainable.”

Buildings and construction together account for 36 per cent of global final energy use and 39 per cent of energy-related carbon dioxide emissions, according to the International Energy Agency’s Global Status Report 2017. This makes sustainable homes the need of the hour.

When it comes to the materials used for constructing the house, you also need to consider their entire lifecycle. “So while a mud block will decompose, a cement block will end up in a landfill,” says Paul, who prefers using mud and lime, and proposes techniques where cement use is significantly reduced.

While you may think wood is sustainable, it being natural, do check if it is sourced ethically, advises Ashwin Alva of Delhi-based Alva Architects. “Ask, is my timber sourced ethically? Is it certified, which means whether it has been legally cut and sourced. These are parameters of environmental safety worldwide.”

Alva also recommends using materials with low VOC—volatile organic compounds—which are neither good for our health nor that of the environment. “Of course, when you are choosing the materials, there is a cost to it, but it is a cost that you are bearing because you are conscious of what you are doing,” he says.

ENERGY EFFICIENT

Making your home energy efficient is critical for your house to be sustainable. One of the first steps is to get the orientation of your house right. “The house needs to be located in such a way that it receives adequate breeze and light,” says Mohanram. For instance, if the house is in a hot and humid city such as Chennai, then you need to ensure that there are adequate windows in the direction of the breeze. “We will, for instance, not locate bedrooms in the West, as they are likely to get hot during the afternoons and evenings. We avoid large openings in the West and East facades and locate them in North and South,” she says. Adds Paul, “An

intelligent layout and reducing the amount of energy-intensive materials such as cement, steel, etc., along with a climate-responsive design will make the house more energy efficient.”

As far as interiors are concerned, it’s best to go for energy-saving appliances. “Something as basic as keeping the air conditioning at 26 degrees helps conserve energy,” says Alva.

“Once we reduce the electricity consumption, the next thing is to power it with solar or wind energy,” says Mohanram. She says for a three-bedroom house you only need 1.5 kW of energy every month. You could put up solar panels on your roof and now there is also technology where parts of the building can be replaced by solar panels. “While such technology



1

An eco-friendly home in Tamil Nadu designed to stay cool without the use of air conditioning

2

Sustainable features like hollow terracotta block walls and exposed brick arches

3

The verandah floats over a water body, collecting rainwater